Exercises: Multivariate Integration

Mathematical Foundations for Machine Learning

Polar Coordinates and Coordinate Transformations

Exercise 1.1 [Easy]
Consider the polar coordinate transformation $\phi(r, \theta) = (r\cos\theta,\ r\sin\theta)^T$.

(a) Compute the Jacobian matrix $D\phi(r, \theta)$.

(b) Compute $\det(D\phi(r, \theta))$ and interpret geometrically what it represents.

Exercise 1.2 [Easy]
Convert the integral $\int\int_D (x^2 + y^2)\, dx\, dy$, where $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq R^2\}$, into polar coordinates. Set up the resulting integral but **do not** evaluate it.

Exercise 1.3 [Medium]
Evaluate the integral

$$I = \int\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\, dx\, dy$$

using polar coordinates. Then use this to show that

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}.$$

We want to show which is the C that normalizesthe

Finally, derive the normalization constant for the univariate Gaussian density $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{y^2}{2\sigma^2}}$.

Exercise 1.4 [Medium]
Consider the transformation $\phi : \mathbb{R}^2_{++} \to \mathbb{R}^2_{++}$ defined by

$$\phi(u, v) = (uv,\ u/v).$$

(a) Compute the Jacobian $D\phi(u, v)$ and its determinant.

(b) Find the inverse mapping $\phi^{-1}(x, y)$. Verify that $\det(D\phi^{-1}(x, y)) = \frac{1}{\det(D\phi(u,v))}$ at corresponding points.

Exercise 1.5 [Hard]
Let $\phi : \mathbb{R}^3_+ \to \mathbb{R}^3$ be the spherical coordinate transformation:

$$\phi(r, \theta, \varphi) = (r\sin\theta\cos\varphi,\ r\sin\theta\sin\varphi,\ r\cos\theta)$$

where $r > 0$, $\theta \in [0, \pi]$, $\varphi \in [0, 2\pi)$.

(a) Compute $\det(D\phi(r, \theta, \varphi))$.

(b) Use the transformation formula and spherical coordinates to evaluate

$$\int\int\int_{\mathbb{R}^3} e^{-\|x\|^2}\, dx$$

and hence show that $\int_{-\infty}^{\infty} e^{-t^2}\, dt = \sqrt{\pi}$ by a purely 3D argument.

(c) Using the result of (b), find the normalization constant for the multivariate Gaussian $\mathcal{N}(0, \sigma^2 I_3)$.

Fubini's Theorem and Iterated Integrals

Exercise 2.1 [Easy]
Let $f(x, y) = x^2 y$ on $K = [0, 2] \times [1, 3]$.

(a) Compute $\int_K f(x, y)\, dx\, dy$ by first integrating over x, then over y.
(b) Compute the same integral by first integrating over y, then over x.
(c) Verify that both orders give the same result. Which theorem guarantees this?

Exercise 2.2 [Medium]
Consider the integral

$$I = \int_0^1 \int_x^1 e^{y^2}\, dy\, dx.$$

(a) Explain why this integral cannot be evaluated directly by first integrating over y.
(b) Sketch the region of integration and swap the order of integration using Fubini's theorem.
(c) Evaluate the resulting integral.

Exercise 2.3 [Medium]
Let $f : \mathbb{R}^3 \to \mathbb{R}$ be defined by $f(x, y, z) = xyz$, and let $K = [0, 1]^3$.

(a) Compute $\int_K f\, d(x, y, z)$ using Fubini's theorem by splitting into three iterated one-dimensional integrals.

(b) Now let $g(x, y, z) = xy + yz$. Compute $\int_K g\, d(x, y, z)$. Explain which property of the integral allows you to split this computation.

Exercise 2.4 [Hard]
In Bayesian machine learning, one often needs to compute the **marginal likelihood** by integrating out a latent variable. Suppose a model has a scalar latent variable w and a scalar observation y, with:

$$p(y \mid w) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-w)^2}{2}}, \qquad p(w) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{w^2}{2\sigma^2}}.$$

Compute the marginal likelihood $p(y) = \int_{-\infty}^{\infty} p(y \mid w)\, p(w)\, dw$ by:
(a) Writing out the full integrand and combining exponents.
(b) Completing the square in w.
(c) Using Fubini/variable substitution to evaluate the resulting Gaussian integral.
State which result from the Gaussian integral computation you are using.

The Transformation Formula

Exercise 3.1 [Easy]
Let $\phi(x) = Ax + b$ where $A \in \mathbb{R}^{n \times n}$ is invertible and $b \in \mathbb{R}^n$. Let $K \subset \mathbb{R}^n$ be compact.

Using the transformation formula, show that

$$\mathrm{vol}(\phi(K)) = |\det(A)| \cdot \mathrm{vol}(K).$$

vol(set) = integral_set(1dx)

Exercise 3.2 [Easy]
Evaluate $I = \int\int_D e^{-(x^2+y^2)}\, dx\, dy$ where $D = \{(x, y) : 1 \le x^2 + y^2 \le 4\}$ is the annular region between circles of radius 1 and 2, using the transformation formula with polar coordinates.

Exercise 3.3 [Medium]
Consider the linear transformation $\phi(u, v) = (2u + v,\ u + 3v)$.

(a) Compute $|\det(D\phi)|$.

(b) Let $S = \{(u, v) : 0 \le u \le 1,\ 0 \le v \le 1\}$ be the unit square. Use the transformation formula to compute

$$\int_{\phi(S)} (x - y)\, dx\, dy.$$

Exercise 3.4 [Medium]
Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be a C^1-diffeomorphism with Jacobian $D\phi(x)$ at point x.

(a) State the relationship between $D(\phi^{-1})(y)$ and $D\phi(x)$ at $y = \phi(x)$.

(b) Prove this relationship using the chain rule applied to $\phi^{-1}(\phi(x)) = x$.

(c) Show that $\det(D(\phi^{-1})(y)) = \frac{1}{\det(D\phi(x))}$ at corresponding points.

Exercise 3.5 [Hard]
The **volume of the n-dimensional unit ball** $B_n = \{x \in \mathbb{R}^n : \|x\|_2 \le 1\}$ can be computed as $V_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)}$.

Using the transformation formula with the substitution $x = ru$ where $u \in \partial B_n$ (the unit sphere) and $r \ge 0$:

(a) Explain why $\mathrm{vol}(B_n(0, R)) = R^n \cdot V_n$ using a simple scaling argument via the transformation formula.

(b) Use part (a) to prove the volume concentration result:

$$\frac{\mathrm{vol}(B_n(0, 1 - \varepsilon))}{\mathrm{vol}(B_n(0, 1))} = (1 - \varepsilon)^n \xrightarrow{n\to\infty} 0$$

for any fixed $0 < \varepsilon < 1$. Give an explicit bound on n such that this ratio is less than δ for given $\delta > 0$.

(c) Discuss the implication for high-dimensional machine learning: why does this mean that data sampled from a uniform distribution on B_n is concentrated near the boundary?

Jacobian of the Inverse

Exercise 4.1 [Easy]
Let $\phi : \mathbb{R} \to \mathbb{R}$ be defined by $\phi(x) = e^x$.

(a) Compute $\phi'(x)$.

(b) Find $\phi^{-1}(y)$ and compute $(\phi^{-1})'(y)$ directly.

(c) Verify that $(\phi^{-1})'(y) = \frac{1}{\phi'(x)}$ at $y = \phi(x)$.

Exercise 4.2 [Medium]
Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $\phi(u, v) = (e^u \cos v,\ e^u \sin v)$.

(a) Compute $D\phi(u, v)$ and show that it is invertible for all (u, v).

(b) Without explicitly finding ϕ^{-1}, compute $D(\phi^{-1})(x, y)$ at $(x, y) = \phi(u, v)$ using the Jacobian-of-the-inverse theorem.

(c) What does the inverse function theorem (Theorem 63) guarantee about ϕ near any point (u_0, v_0)?

Exercise 4.3 [Hard]
In normalizing flows (a generative ML model), one transforms a simple base distribution $p_Z(z)$

(e.g., Gaussian) through an invertible map $f : \mathbb{R}^d \to \mathbb{R}^d$ to obtain a complex distribution $p_X(x)$ for $x = f(z)$.

The **change of variables formula** for probability densities gives:

$$p_X(x) = p_Z(f^{-1}(x)) \cdot |\det(D(f^{-1})(x))|.$$

(a) Using the Jacobian-of-the-inverse theorem, rewrite this as:

$$p_X(x) = p_Z(z) \cdot |\det(Df(z))|^{-1}$$

where $z = f^{-1}(x)$. Justify each step.

(b) Suppose f is a composition of K invertible maps: $f = f_K \circ f_{K-1} \circ \cdots \circ f_1$. Show that

$$\log p_X(x) = \log p_Z(z_0) - \sum_{k=1}^{K} \log |\det(Df_k(z_{k-1}))|$$

where $z_0 = f^{-1}(x)$ and $z_k = f_k(z_{k-1})$.

(c) Explain why computing $\det(Df_k)$ for each layer is the computational bottleneck and why architectures like RealNVP use transformations with **triangular** Jacobians.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Let $A \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix with eigendecomposition $A = Q\Lambda Q^T$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_i > 0$.

Consider the integral

$$I = \int_{\mathbb{R}^n} e^{-\frac{1}{2}x^T Ax}\, dx.$$

(a) Apply the transformation formula with $x = Qy$ (where Q is orthogonal). What is $|\det(DQ)|$? Show that the integral becomes $\int_{\mathbb{R}^n} e^{-\frac{1}{2}y^T \Lambda y}\, dy$.

(b) Use Fubini's theorem to split this into n independent one-dimensional Gaussian integrals and show that

$$I = \frac{(2\pi)^{n/2}}{\sqrt{\det(A)}}.$$

(c) Use this result to verify that the multivariate Gaussian $\mathcal{N}(\mu, \Sigma)$ density

$$p(x) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$$

integrates to 1.

Exercise 5.2 [Hard]
(Change of variables for probability densities) Let $Z \sim \mathcal{N}(0, I_2)$ and define the transformation $\phi(z_1, z_2) = (z_1, z_1 + z_2)$.

(a) Compute $D\phi$ and $|\det(D\phi)|$.

(b) Find $\phi^{-1}(x_1, x_2)$ and compute $D(\phi^{-1})$ both directly and via the Jacobian-of-the-inverse theorem.

(c) Using the change of variables formula for densities, find $p_X(x)$ where $X = \phi(Z)$.

(d) Identify the distribution of $X = (X_1, X_2)^T$. What is its covariance matrix? Verify by computing $\mathrm{Cov}(X_1, X_2) = \mathrm{Cov}(Z_1, Z_1 + Z_2)$ directly.

Exercise 5.3 [Hard]
(Expectation via the transformation formula) Let $X \sim \mathcal{N}(0, \sigma^2)$ be a univariate Gaussian. Using polar coordinates in $\mathbb{R}^2$ and Fubini's theorem:

(a) Show that $\mathbb{E}[X^2] = \sigma^2$ by computing

$$\mathbb{E}[X^2] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2\sigma^2}}\, dx$$

via the trick of writing $\mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]$ for i.i.d. copies X, Y, converting to polar coordinates, and factoring.

(b) More generally, show that for $X \sim \mathcal{N}(0, \sigma^2 I_n)$, the transformation formula and Fubini give $\mathbb{E}[\|X\|^2] = n\sigma^2$.